

TOO SMALL (< 3-inch wheelbase)

Cannot carry Pi 5
companion
computer (~60 g +
peripherals)

Cannot carry protective
cage
(PETG, 35.8 cm Ø, bearing
assembly)

Insufficient TWR margin for
payload overhead (< 2:1)

SWEET SPOT: 4-INCH

✓ Carries Pi 5 + cage +
sensors

✓ TWR ~3.8:1 (hover at
~45-50%)

✓ 2004 motors → ~860g
thrust/motor

✓ 6S battery → efficient
mid-throttle

✓ Upper indoor safety limit

TOO BIG (>5-inch / LIDAR-class)

Exceeds indoor safety
envelope

Too heavy for confined-
space deployment

Propeller wash destabilises
in tight enclosures

LIDAR payload (e.g.
Ouster OS0)
adds ~450 g — exceeds
TWR budget